PA-07 Bellwether Precincts: Full Results

This memo brings together every bellwether test I ran. It covers how the precincts were chosen, how stable the choice is, what their registration and demographics look like, how well the baseline model predicts the district from these precincts alone, what that implies for 2026, and how the 10-precinct set compares with Professor Crain's 6-precinct set when both go through the same tests.

Summary of findings

1. **A fixed rule finds 10 precincts that vote like the district.** Combined, they were within 0.2 points of the district's Democratic share in both 2022 and 2024.
2. **The set is stable; individual precincts are not.** Chosen on 2022 alone, the set missed 2024 by 0.57 points, and 9 of 10 precincts stayed within 2 points. Only 5 of those 10 remain once 2024 data are added, so the combined vote of the set is the signal, not any one precinct.
3. **The baseline model, run on these 10 precincts alone, predicted 2024 within a third of a point** (0.14 points after adjusting for the set's usual lean) and called the winner correctly.
4. **For 2026, the bellwethers point to a toss-up.** The baseline forecasts these precincts at 50.0% for Brooks, while its full district forecast is 48.7%. Most of that gap comes from the full forecast's assumption that 2026 turnout will look like 2022's.
5. The 10-precinct set and Professor Crain's 6-precinct set share no precincts and pass different tests**.** The 10-precinct set fits the current district better in level, swing and the 2024 backtest; the 6-precinct set holds up better under the old district lines. Adjusted for their lean, the sets point to 50.0% and 49.2% for Brooks, both closer than the full baseline.

Key terms

- **Democratic share:** Democratic House votes ÷ (Democratic + Republican House votes). Third-party votes are excluded.
- **Gap:** a precinct's Democratic share minus the district's, in percentage points. Zero means it voted exactly like the district.
- **Combined share:** the 10 precincts' votes added together and treated as one precinct.
- **Offset adjustment:** subtracting a set's usual lean from its result, so a set that normally runs 1 point Democratic is read 1 point lower.
- **Out of sample:** tested on an election that was not used to choose the precincts.

- **Net registration margin:** (registered Democrats − registered Republicans) ÷ all registered voters. It measures party sign-ups, not votes.
- **Baseline:** the group's precinct-level random forest forecast (v0.1), which forecasts the district at 48.7% for Brooks.

1. Selection rule

The rule was written before looking at which precincts it would choose, and it is applied by formula. No precinct was added or removed by hand.

1. **Universe:** all precincts in PA-07 as currently drawn: Carbon, Lehigh and Northampton, plus the Eldred, Polk and Ross (USC 7) precincts of Monroe.
2. **Size:** at least 300 two-party House votes in both 2022 and 2024. In smaller precincts, a few voters can move the share by a point or more.
3. **Still exists in 2026:** precincts in the seven Northampton towns whose lines were redrawn for 2026 are excluded, because they cannot be observed on election night. 296 precincts qualify.
4. **Score:** the larger of the precinct's 2022 gap and 2024 gap, ignoring sign. A bellwether has to be close in both elections, not close on average.
5. **Selection:** the 10 precincts with the lowest scores. Ties go to the precinct with more votes.

2. The bellwether set

Precinct	County	Gap 2022	Gap 2024	Net registration, 2026	Baseline forecast 2026 (Brooks)
Salisbury Ward 2	Lehigh	+0.08	+0.04	+9.1%	50.1%
Salisbury Ward 5, District 2	Lehigh	+0.37	+0.41	−2.6%	50.2%
Lower Macungie District 2	Lehigh	+0.58	+0.35	−3.8%	50.9%

Salisbury Ward 1	Lehigh	−0.83	−0.41	+8.8%	49.0%
Upper Saucon District 1	Lehigh	−0.15	−0.86	−8.0%	49.4%
Nazareth Borough Ward 1	Northampton	−0.93	+0.34	−6.9%	49.3%
North Catasauqua Ward 2	Northampton	+0.98	+0.99	+3.7%	50.7%
Lower Macungie District 10	Lehigh	−0.29	−1.06	−6.9%	49.2%
Lower Macungie District 1	Lehigh	+1.09	+0.07	−1.0%	50.6%
Northampton Borough Ward 2	Northampton	−0.26	−1.14	+4.2%	48.9%
Combined		**+0.16**	**−0.13**	**−1.0%**	**50.0%**

The district was 50.98% Democratic in 2022 and 49.50% in 2024. Together the 10 precincts cast about 17,500 two-party votes in 2024, roughly 4% of the district.

3. Stability tests

Out of sample. The same rule applied to 2022 alone chose 10 precincts. In 2024, which played no part in choosing them, 9 of 10 stayed within 2 points of the district, and the set's combined vote missed the district by 0.57 points.

Carryover. Only 5 of those 10 precincts remain in the set once 2024 is added. Individual precincts drift in and out; the combined vote of the set is what holds.

Dropping precincts. Removing any one precinct leaves the set's 2024 gap between −0.25 and −0.01 points. The worst-fitting precinct, Northampton Borough Ward 2 (largest score, 1.14), changes the 2024 gap from −0.13 to −0.05 points when removed. No single precinct carries the result.

Old district lines. In 2018 and 2020, under the old PA-07 (55.2% and 51.9% Democratic, without Carbon), these precincts ran 0.8 to 6.6 points more Republican than the district, about 4 points on average. They are bellwethers for the current district only.

Geography. All 10 are suburban precincts in Lehigh and Northampton. None are in Carbon, Allentown, Bethlehem or Easton. They match the district's overall result, not its mix of places.

4. Registration trend

Net registration margin (D − R)	2022	2024	2026 primary	Change 2024→2026
Bellwether set	+0.9%	−1.3%	−1.0%	+0.4 pts toward D
District	+7.8%	+4.5%	+3.3%	−1.2 pts toward R
Set minus district	−6.9 pts	−5.8 pts	−4.3 pts	

Two patterns stand out:

- **These precincts vote like the district while being registered 4 to 7 points less Democratic.** Democratic House candidates do better here than registration alone would suggest.
- **Since 2024 the set has edged toward Democrats while the district moved toward Republicans.** The gap between them has narrowed every cycle.

The latest figures use registration as of the May 2026 primary. They should be updated with the October 5 district snapshot.

5. Demographics

The county voter files record each voter's birth year, sex, party and whether (never how) they voted.

2024 general election	Bellwether set	District
Registered voters who voted	81.6%	75.7%
Voters registered Republican	43.7%	41.9%
Voters registered Democratic	40.5%	41.5%
Registered voters Baby Boomer or older	38.9%	34.7%
Registered voters in Gen Z	12.4%	14.7%

The bellwether electorate is older, turns out more, and is more Republican by registration than the district's, yet it votes like the district. That means unaffiliated voters and some Republicans here vote Democratic for House more often than elsewhere. So the set reflects the district's result, not its make-up. Because older, high-turnout electorates drop off less in midterms, the district's 2026 electorate may look more like this set's than it did in 2024.

Turnout figures come from the August 31, 2026 voter file, which misses 2024 voters who have since left the roll. The comparison between set and district is more reliable than either number alone.

6. Backtest: the baseline run on the bellwethers alone

The question: if we used the baseline model's predictions for only the bellwether precincts, and read their combined share as the district's, how close would we get?

The main test uses the 10 precincts chosen from **2022 results only**, so 2024 played no part in choosing them.

Method	Estimate	Actual 2024	Miss	Correct winner?
Baseline on bellwethers chosen from 2022	49.18%	49.50%	−0.31 pts	Yes
Same, offset-adjusted	49.36%	49.50%	−0.14 pts	Yes
Baseline on current bellwethers (chosen from 2022 and 2024)	49.73%	49.50%	+0.23 pts	Yes
Same, offset-adjusted	49.57%	49.50%	+0.07 pts	Yes
Full baseline, all precincts	49.40%	49.50%	−0.10 pts	Yes
For reference: repeat the 2022 result	50.98%	49.50%	+1.49 pts	No

Run on just 10 precincts, about 4% of the vote, the baseline predicted the district within a third of a point. With the offset adjustment it came within 0.14 points, close to the full model's 0.10.

One caution: the baseline's 2024 backtest counted votes using actual 2024 turnout, so this tests vote share, not turnout. It is also one election.

7. 2026 forecast from the bellwethers

Method	Brooks share	Implied result

Baseline on current bellwethers	49.99% (7,040 to 7,044 votes)	Toss-up
Same, offset-adjusted	49.97%	Toss-up
Baseline on bellwethers chosen from 2022	49.69% (offset-adjusted 50.06%)	Toss-up
Full baseline, all precincts	48.73%	Mackenzie by about 2.5 pts

Read through the bellwethers, the baseline points to a race at essentially 50–50. Read across all precincts, the same model gives Mackenzie a 2.5-point win.

Why they differ. The full forecast converts each precinct's predicted share into votes using its 2022 midterm turnout. The bellwether reading does not depend on turnout elsewhere in the district. When I reweighted the baseline's own 2026 precinct shares by 2024 turnout instead of 2022 turnout, the forecast moved about 0.8 points toward Brooks (tested on the 280 precincts that match across files). So roughly two-thirds of the 1.3-point gap comes from the turnout assumption, the one part of the model the 2024 backtest never tested.

8. Side by side: the 10-precinct set and Professor Crain's 6-precinct set

The two sets share no precincts. One of Professor Crain's precincts, Lower Macungie District 3, appeared in the 10-precinct set chosen from 2022 alone but dropped out when 2024 was added.

They also define a bellwether differently:

- **The 10-precinct set** is chosen for **level**: precincts that vote at nearly the same Democratic share as the district.
- **The 6-precinct set** is chosen for **correlation**: precincts that rise and fall with a 200-precinct benchmark, even if they sit well above or below it. Its precincts sit between 2.7 and 18.4 points from that benchmark.

Both sets were run through the same tests below. District values are shown where they apply.

Fit to the district

Test	10-precinct set	6-precinct set
Combined gap, 2022	+0.16 pts	+2.22 pts
Combined gap, 2024	−0.13 pts	+1.36 pts
Swing 2022→2024 (district −1.49)	−1.78 pts	−2.34 pts
Average swing miss per precinct, 2022→2024	0.54 pts	1.26 pts
Combined gap, 2018, old lines	−4.28 pts	−0.74 pts
Combined gap, 2020, old lines	−3.12 pts	−1.02 pts
Swing 2018→2020, old lines (district −3.30)	−2.14 pts	−3.58 pts
2024 two-party votes	about 17,500	about 11,000

Stability

Test	10-precinct set	6-precinct set
Change in lean, 2022→2024	−0.29 pts	−0.86 pts

2024 gap with any one precinct removed	−0.25 to −0.01 pts	−0.41 to +3.09 pts
Change in lean with any one precinct removed	−0.40 to −0.19 pts	−1.08 to −0.77 pts
Swing with worst-fitting precinct removed	−1.89 pts (Nazareth Borough Ward 1 removed)	−2.26 pts (Upper Macungie District 4 removed)
Out-of-sample test (chosen on 2022, scored on 2024)	Missed by 0.57 pts	Not possible; the data used to choose the set are not available

The change in lean matters because a correlation-based set has to be read with an offset adjustment. If the set's lean shifts between elections, the adjusted reading misses by about that amount. Removing Bethlehem 9th Ward East moves the 6-precinct set's level by several points, but not its swing.

Registration and demographics

Measure	10-precinct set	6-precinct set	District
Net registration margin, 2022	+0.9%	+1.7%	+7.8%
Net registration margin, 2024	−1.3%	−0.2%	+4.5%
Net registration margin, 2026 primary	−1.0%	−0.1%	+3.3%

Change since 2024	+0.4 pts toward D	+0.1 pts toward D	−1.2 pts toward R
Registered voters who voted, 2024	81.6%	81.0%	75.7%
2024 voters registered Republican	43.7%	43.0%	41.9%
2024 voters registered Democratic	40.5%	40.4%	41.5%
Registered voters Baby Boomer or older	38.9%	36.9%	34.7%
Registered voters in Gen Z	12.4%	13.6%	14.7%

Baseline model run on each set

Test	10-precinct set	6-precinct set
2024 estimate, read directly (actual 49.50%)	49.73% (miss +0.23)	51.09% (miss +1.60; wrong winner)
2024 estimate, offset-adjusted	49.57% (miss +0.07)	48.88% (miss −0.62)
2026 forecast for the set (Brooks)	49.99%	51.00%

2026 forecast, offset-adjusted	49.97%	49.21%
Full baseline, all precincts, 2026	48.73%	48.73%

What the comparison shows

- **Under the current lines, the 10-precinct set fits better.** It is closer to the district in both level and swing for 2022–2024, its lean barely moved, and the baseline run on it came within 0.1 points of the 2024 result.
- **Under the old lines, the 6-precinct set fits better.** It tracked the 2018→2020 swing within 0.3 points and stayed within about 1 point of the old district, while the 10-precinct set ran 3 to 4 points more Republican. This fits a set chosen on correlation across many elections: it holds up across a redistricting, which the 10-precinct set does not.
- **The 6-precinct set must be read with an offset adjustment.** Read directly, it would have called 2024 for the wrong candidate. Its lean shifted 0.86 points between 2022 and 2024, which accounts for most of its 0.62-point adjusted miss.
- **The two are demographically similar.** Both are older and higher-turnout than the district and more Republican by registration, yet both vote close to it. The 6-precinct set is slightly closer to the district's age mix.
- **Both point to a closer 2026 race than the full baseline.** The adjusted readings are 49.2% (6 precincts) and 50.0% (10 precincts), against 48.7% for the full baseline.

On election night the two sets work best together. If both come in near their forecast shares, the closer-race reading is supported from two independent directions.

Note: "Bethlehem City, Ward 9, District 2" on the original list was matched to the precinct the baseline files call Bethlehem 9th Ward East. Its distance from the district (about 18 to 20 points) matches the listed 18.4 points. This match should be confirmed.

9. Limitations and next steps

1. **Two elections.** Only 2022 and 2024 use the current lines, and neither was a midterm under these lines with a Republican president.
2. **The baseline's backtest used actual turnout.** Its accuracy on vote counts, and so the 48.7% headline, depends on a turnout assumption that has not been tested.
3. **October 5 snapshot.** The registration figures should be updated with the October 5 district snapshot.

4. **Election night.** Compare the 10 precincts' returns with the baseline's 50.0% forecast for them. Near 50%, the baseline's precinct picture holds. Well below it, the full 48.7% forecast is more likely right.

Sources

1. Pennsylvania Department of State, Voting and Election Statistics: precinct registration, 2022, 2024 and the May 2026 primary.
2. Pennsylvania Department of State, Election Data: certified U.S. House results by precinct, 2018–2024. Precinct totals match the certified district totals of 296,891 (2022) and 403,314 (2024) two-party votes.
3. Muller, Max. "Consolidated and Reorganized Data," PA_07_Precinct_Data.xlsx (Precinct Summary, Precinct Consolidated and Trends tabs). PA-07 Project Tracker, September 17, 2026. Compiles sources 1 and 2 and the county voter files (August 31, 2026 snapshot).
4. PA07_2026_Forecast_Precinct.xlsx, the group's baseline forecast (random forest v0.1): 2026 Precinct Forecast and 2024 Backtest tabs.
5. Professor Crain's bellwether precinct list (correlation with a 200-precinct benchmark), shared October 2026.